Advection Assignment

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1 Introduction

The aim of this project is to find and test advection schemes that perform better than the simplest explicit, linear finite difference schemes and make recommendations of schemes that may be suitable for weather and climate models. Theoretical properties of the alternatives are demonstrated through a set of one-dimensional numerical experiments and are compared with the simple schemes.

2 Basic Schemes

We define schemes for solving the one-dimensional linear advection equation for ϕ:

$$\frac{\partial \phi}{\partial t} = -u\frac{\partial \phi}{\partial x}, \tag{1}$$

on a periodic domain $x \in [0, 1)$ with constant wind speed, u, on a grid of n_x evenly spaced points a distance Δx apart with n_t time steps of length Δt. Grid points are labelled by j and time steps are labelled by n and $\phi_j^{(n)}$ is ϕ at grid point j and time level n. The Courant number is defined as $c = u\Delta t/\Delta x$.

2.1 Forward in Time, Backward in Space (FTBS)

The forward in time, backward in space (FTBS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n)} - c(\phi_j^{(n)} - \phi_{j-1}^{(n)}). \tag{2}$$

FTBS is first-order accurate in space and time (Weller, 2025). It is stable for $c \in [0, 1]$. Rearrangement of (2) shows that, for $c \in [0, 1]$, $\phi_j^{(n+1)}$ is a convex combination of $\phi_j^{(n)}$ and $\phi_{j-1}^{(n)}$, so FTBS is bounded for $c \in [0, 1]$:

$$\phi_j^{(n+1)} = (1 - c)\phi_j^{(n)} + c\phi_{j-1}^{(n)}. \tag{3}$$

2.2 Forward in Time, Centred in Space (FTCS)

The forward in time, centred in space (FTCS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n)} - \frac{c}{2}(\phi_{j+1}^{(n)} - \phi_{j-1}^{(n)}). \tag{4}$$

FTCS is second-order accurate in space and first-order in time but is unconditionally unstable (Weller, 2025). Results of FTCS are shown in section 3 but only to help to understand the CTCS results.

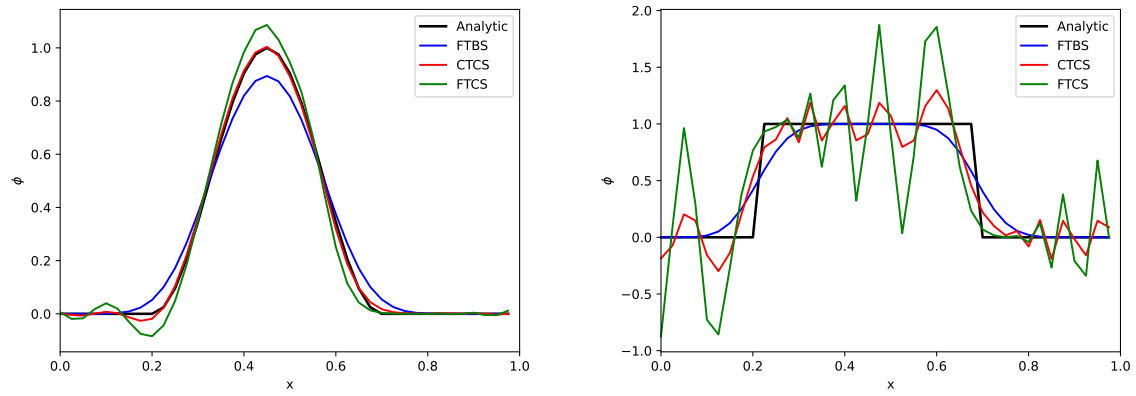


Figure 1: Solution of the advection equation in comparison to the analytic solution for initial conditions (6) and (7).

2.3 Centred in Time, Centred in Space (CTCS)

The centred in time, centred in space (CTCS) finite difference scheme is:

$$\phi_j^{(n+1)} = \phi_j^{(n-1)} - c(\phi_{j+1}^{(n)} - \phi_{j-1}^{(n)}). \tag{5}$$

CTCS is second-order accurate in space and time, for the same cost as FTBS, and is stable for $c \in [-1, 1]$ (Weller, 2025). CTCS is a three time-level scheme so requires more storage. There is also a risk that solutions at adjacent times and grid points can display spurious behaviour.

3 Results of the Basic Schemes

Solutions are shown starting from two sets of initial conditions:

$$\phi(x,0) = \begin{cases} \frac{1}{2}\left(1 - \cos 2\pi \frac{x-a}{b-a}\right) & \text{for } a \le x < b \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

$$\phi(x,0) = \begin{cases} 1 & \text{for } a \le x < b \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

with $a = 0$ and $b = 0.5$.

Figure 1 shows results after 20 time steps using a Courant number of 0.4 and 40 points in space in comparison to the analytic solution.

3.1 Spurious Oscillations

Discontinuities clearly pose a significant obstacle for numerical schemes as shown for (7). Both CTCS and FTCS exhibit spurious oscillations which appear particularly strong just upstream (where our "flow" travels in the positive x direction) of the two discontinuities. We can even get a glimpse of these issues for (6), as small oscillations can be seen for FTCS just upstream of $x = 0.2$: this is where the second derivative of our exact solution is discontinuous. We might conjecture that a similar oscillation arises just upstream of $x = 0.8$, but that they are masked by the curve of the cosine. These oscillations arise from the fact that FTCS and CTCS are second order schemes and thus cannot be monotonous, as described in Godunov and Bohachevsky (1959).

3.2 Diffusivity

In neither case does FTBS display these oscillations, but instead, the solutions appear to smooth our initial condition. Indeed, if we expand (2) as a Taylor series around $(j\Delta x, n\Delta t)$, we get

$$\phi + \Delta t\phi_t + \frac{1}{2}\Delta t^2\phi_{tt} + \mathcal{O}(\Delta t^2) = (1-c)\phi + c\left(\phi - \Delta x\phi_x + \frac{1}{2}\Delta x^2\phi_{xx}\right) + \wr(\Delta x^2).$$

We can use the definition of the Courant number and (1) to simplify this expression to:

$$\begin{aligned}\phi_t + u\phi_x &= \frac{1}{2}\Delta tc(1-c)\phi_{xx} + \mathcal{O}(2)\\ &= D\phi_{xx} + \mathcal{O}(2),\end{aligned}$$

where the numerical diffusivity is given by $D = \Delta tc(1-c)$. From this, we observe:

- $D \propto \Delta t$. Therefore, reducing the time-step reduces the numerical diffusion but at a computational cost.
- For a fixed time step, numerical diffusion is minimised when $c \approx 0$ or 1 and maximised when $c = 0.5$. Therefore, for our $c = 0.4$, we expect to see a significant amount of numerical diffusion.
- Discontinuities produce high frequency Fourier modes which decay exponentially at a rate proportional to their frequency squared. This explains the much greater level of diffusion for initial condition (7).

4 Alternative Scheme

4.1 Motivation

Consider a general forward in time scheme, quasi-centred in space, given by:

$$\phi_j^{n+1} = \alpha\phi_{j-1}^n + \beta\phi_j^n + \gamma\phi_{j+1}^n, \tag{8}$$

where α, β and γ may be varied.

To ensure consistency to first order, we require that

$$\alpha + \beta + \gamma = 1 \tag{9a}$$

$$\alpha - \gamma = c \tag{9b}$$

This leaves us with one degree of freedom.

Inspired by the conserved quantities of the advection equation, suppose we wanted some general quantity S_n to be conserved by the scheme. For our purpose, we can define this quantity somewhat generally in terms of some inner product so that $S_n = \langle\phi^n, \phi^n\rangle$ where our ϕ^n's are now vectors.

In order for this inner product to make sense for our purpose, we restrict to those for which translational symmetry holds, i.e.

$$\langle\mathcal{T}a, \mathcal{T}b\rangle = \langle a, b\rangle \tag{10}$$

where $\mathcal{T}$ is the discrete translation operator around the torus satisfying $(\mathcal{T}a)_j = a_{j-1}$.

Now we choose the coefficients of our scheme to ensure that $S^{n+1} = S^n$, i.e. conservation of S.

4.2 Deriving Coefficients

We begin by expanding an expression for S_{n+1}:

$$\begin{aligned} S^{n+1} &= \langle \phi^{n+1}, \phi^{n+1} \rangle \\ &= (\alpha \mathcal{T} \phi^n + \beta \phi^n + \gamma \mathcal{T}^{-1} \phi^n, \alpha \mathcal{T} \phi^n + \beta \phi^n + \gamma \mathcal{T}^{-1} \phi^n \rangle \\ &= (\alpha^2 + \beta^2 + \gamma^2) S^n + 2\beta(\alpha + \gamma) \langle \mathcal{T} \phi^n, \phi^n \rangle + 2\alpha\gamma \langle \mathcal{T}^2 \phi^n, \phi^n \rangle \end{aligned}$$

Setting $S_{n+1} = S_n$, we obtain the equation

$$(\alpha^2 + \beta^2 + \gamma^2 - 1) \langle \phi^n, \phi^n \rangle + 2\beta(\alpha + \gamma) \langle \mathcal{T} \phi^n, \phi^n \rangle + 2\alpha\gamma \langle \mathcal{T}^2 \phi^n, \phi^n \rangle = 0 \tag{11}$$

Once we have calculated the inner products in this equation, we can solve the nonlinear equations (9) and (11) to obtain the coefficients α, β and γ.

For this project, we will conserve the integral of the function squared, i.e. $\int \phi^2 dx \approx \frac{1}{\Delta x^2} \sum (\phi_i)^2$ and so define $S^n = \sum (\phi_i^n)^2$.

4.3 Theoretical Properties

4.3.1 Limiting Oscillations

Ignoring numerical errors, this scheme will ensure that

$$\sum (\phi_i^n)^2 = \Delta x^2 D^2 \tag{12}$$

where D is a constant independent of n which can be calculated from the initial value ϕ_0^n. Since all terms in the sum are positive, we can deduce that

$$|\phi_i^n| \leq \Delta x D \tag{13}$$

for all time, giving us stability. Note: this scheme is universally stable for any c, allowing us to take large time steps.

Now suppose we Fourier decompose a general function f defined on the unit torus so

$$f(x) = \Re \left\{ \sum_{k=0}^{\infty} \hat{f}_k e^{2\pi i k x} \right\}. \tag{14}$$

Then

$$\int_0^1 f(x)^2 dx = \sum_{k=0}^{\infty} |\hat{f}_k|^2. \tag{15}$$

We can view (15) as a constraint on the magnitude of oscillation terms, thus acting as a dampener to spurious oscillations.

4.3.2 Limiting Diffusion

Consider the advection-diffusion equation

$$\partial_t \phi + u \partial_x \phi = D \partial_x^2 \phi \tag{16}$$

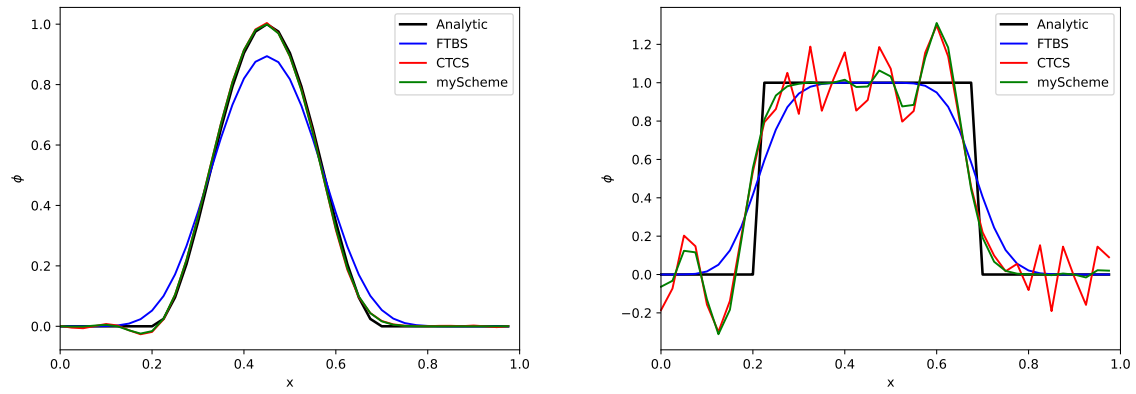


Figure 2: Solution of the advection equation in comparison to the analytic solution for initial conditions (6) and (7) for the scheme defined in section 4.

Here D represents our unwanted numerical diffusion. Under this scheme, we can calculate how our $S = \int_0^1 \phi^2 dx$ might naturally evolve:

$$\begin{aligned}\frac{d}{dt}\int_0^1 \phi^2 dx &= 2\int_0^1 \phi\phi_t dx \\ &= 2D\int_0^1 \phi\phi_{xx} dx \\ &= -2D\int_0^1 \phi_x^2 dx,\end{aligned}$$

where we have used integration by parts to obtain the final line. From this, we see that $\frac{dS}{dt}$ is always negative and so S decreases until we reach a steady state, i.e. a constant ϕ.

From this, we glean two points of note:

- S provides a good metric for the amount of numerical diffusion.
- Our new scheme should minimise this diffusion by preventing the decay of S.

5 Results

5.1 Snapshots of the Solutions

The results of the new scheme for the setup defined in section 3 are shown in figure 2 which are compared with CTCS and FTBS.

For both initial conditions, we find that our new scheme appears to follow the CTCS scheme quite closely whilst requiring only a single timestep. We might expect that this is due to CTCS naturally doing quite a good job at conserving S. Whilst this might suggest CTCS should be prioritised given the increased complexity of our algorithm, we should also remember that single time-step schemes use much less storage than two-step schemes, and thus our choice might be determined by whether storage or computational power are our limiting considerations.

As hoped, for initial condition (6), our scheme avoids the diffusion seen with FTBS, instead matching much closer to our analytic solution. A similar result occurs for initial condition (7), but in this case, we trade a reduction in diffusion with an increase in spurious oscillations.

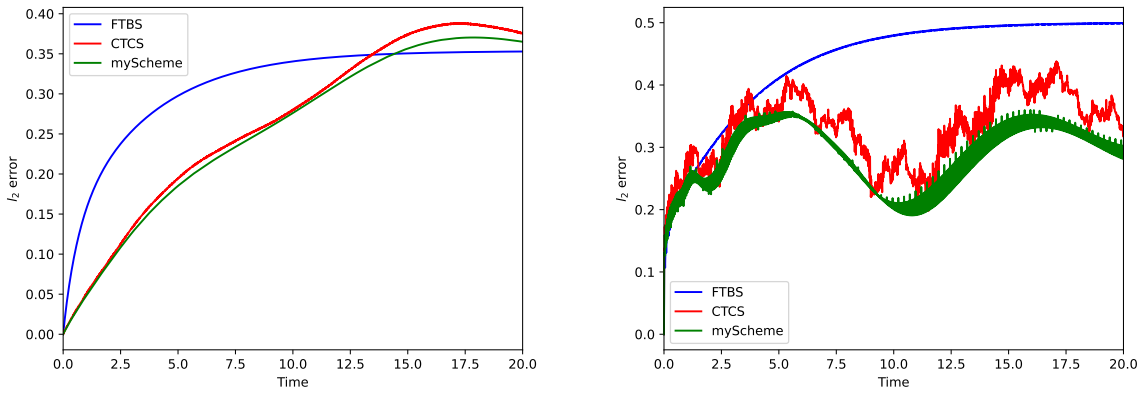


Figure 3: Growth of l_2 error from $t = 0$ to 20 for initial conditions (6) and (7) for the scheme defined in section 4.

Godunov and Bohachevsky (1959) describe this trade off for second-order spatial schemes, but it is interesting that this extends to our scheme which has only been constructed to first order. It appears that, by closely approximating the second-order CTCS, a similar phenomenon has arisen.

Finally, our scheme appears to have been successful in reducing the amount of spurious oscillations in initial condition (7) as compared to the CTCS scheme. This is particularly clear when looking in $x \in [0.3, 0.6] \cup [0.7, 1]$. It appears that, for both our new scheme and CTCS, the largest oscillations arise just upstream of the discontinuity, but unlike CTCS, our these oscillations are damped much faster as we move further upstream.

Figure 3 displays the l_2 error of each scheme over a longer time scale. From this we see that each scheme has errors of similar orders of magnitude in each case. Whilst the new scheme largely outperforms the other two schemes, this difference is not large enough to justify the use of any one scheme without further consideration. In particular, note that the diffusion present in FTBS leads to the error grow quickly to start and then level off rapidly as the numerical solution converges to a constant function. From one perspective, this is valuable as it introduces a hard limit on the l_2 error, but we also lose all information about the initial conditions. We need further tests to evaluate these different schemes.

5.2 Conservation of S and Other Quantities

Since our initial motivation for our new scheme was conserving the quantity $S_n = \sum_{i=0}^{N_x} (\phi_i^n)^2$, we will first check that S is indeed conserved. To do, we will plot for each scheme and for each initial condition, the magnitude of the difference between the initial value of S against value of S calculated at time t. These results are shown in figure 4.

These results indeed show that the new scheme conserves S to a high degree of accuracy, of orders of magnitude $\approx 10^{-14}$. Note, this value is not conserved exactly for two reasons:

- Numbers stored by computers have a finite precision, so an exact solution to (11) cannot be stored, giving us some unavoidable errors.
- At each point we set $S_n - S_{n-1} = 0$ and so any computational error in choosing our coefficients will propagate forwards. This can be seen in the plot for initial condition (6)

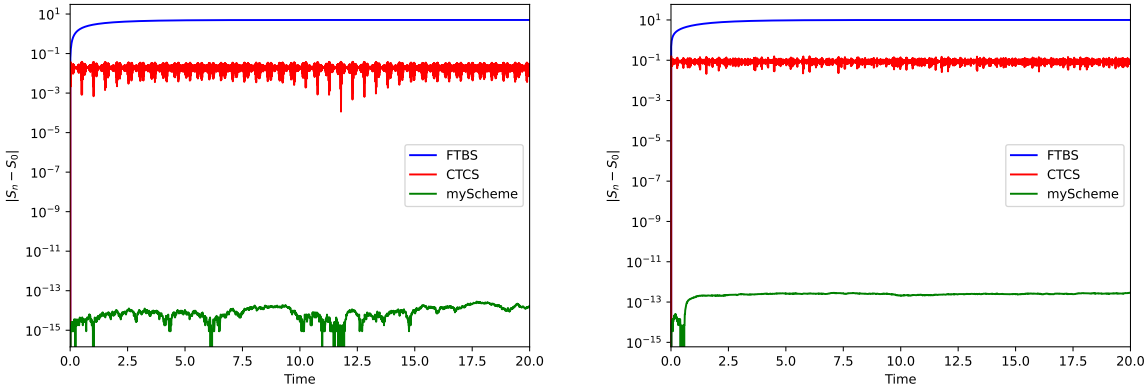


Figure 4: Change in $S = \sum \phi_i^2$ from its inital value for initial conditions (6) and (7) for the scheme defined in section 4.

where the error increases from 10^{-15} to 10^{-14}, although this increase could equally be noise. [1]

From this plot we also see that the S-error grows rapidly to become of an order comparable to the initial conditions. This is to be expected from our considerations of diffusion in section 4.3.2.

Less expected is how the CTCS scheme has successfully limited the S-error to a magnitude much smaller than the diffusive FTBS. Although this pales in comparison to our new scheme, it suggests that CTCS does a good job at conserving this quantity without being specifically designed to do so.

These results are largely to be expected since we have designed the new scheme for one specific purpose: conserving S. We might, however, also be interested to see how our scheme performs for conserving different inner products as defined in section 4.1.

One physically relevant inner product might be the following:

$$\langle f, g \rangle = \sum_{i=1}^{N_x} (f_i - f_{i-1})(g_i - g_{i-1}), \tag{17}$$

so that our new conserved quantity, dS say, is given by

$$dS = \sum_{i=1}^{N_x} (\phi_i - \phi_{i-1})^2. \tag{18}$$

One physical example of where this norm might be useful is if ϕ is the velocity potential of a flow - then dS is proportional to the total kinetic energy of the system. We carry out the test from figure 4 in figure 5 but instead calculating the error $dS_n - dS_0$ at each time step.

From the plots in figure 5, we see that whilst our new scheme does an exceptional job at conserving S, it performs much worse at conserving dS. That said, it still performs much better than FTBS.

The biggest surprise from this test was how consistently CTCS conserved both S and dS. Moreover, neither of these errors appear to grow over time, instead maintaining a constant

[1] An easy fix to this issue would be calculating S_0 and then setting $S_n - S_0 - 0$ at each step. This has the added benefit of halving the computations calculating S values at each step.

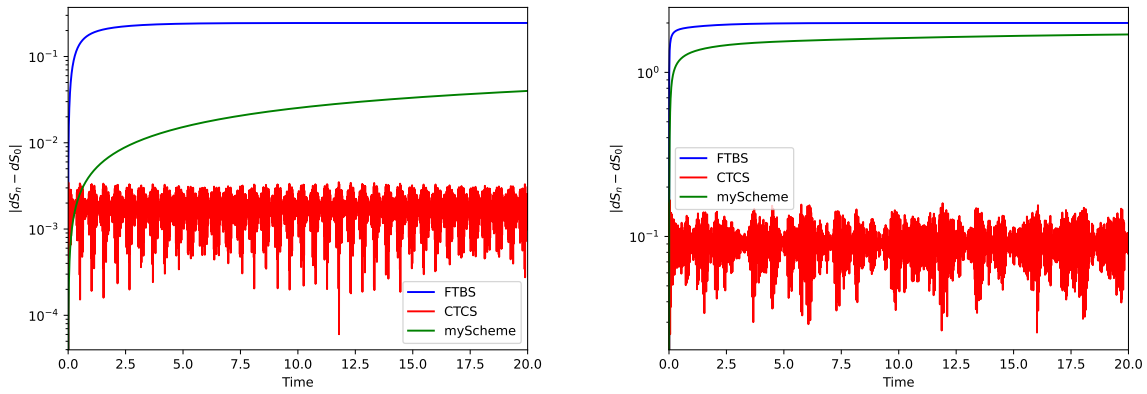


Figure 5: Change in $dS = \sum(\phi_i - \phi_{i-1})^2$ from its inital value for initial conditions (6) and (7) for the scheme defined in section 4.

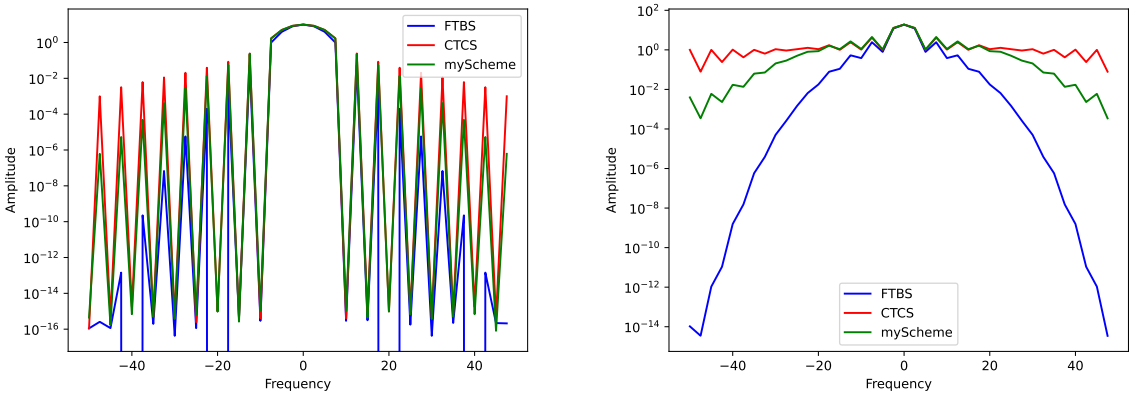


Figure 6: Fourier transform of the numerical solution at $t = 0.2$ for initial conditions (6) and (7) for the scheme defined in section 4.

magnitude. This suggests that, whilst our scheme may be optimal if we wish to conserve just one value as precisely as possible, CTCS naturally performs better at conserving several physically relevant values at the same time.

5.3 Fourier Analysis of Oscillations

We expected that our new scheme might also reduce spurious oscillations, and so to verify this, we calculate the Fourier transform of our numerical solutions at $t = 1$, which will show us the amplitude of oscillations at different frequencies. These results are shown in figure 6.

As expected, the Fourier coefficients for FTBS decay rapidly as the frequency increases as diffusion cause these modes to decay exponentially with time and frequency. Meanwhile, particularly for initial condition 7, CTCS appears to maintain oscillations of all frequencies, contributing to the spurious oscillations observed in figure 2. The new scheme appears to strike a balance between these two extremes: oscillations of higher frequencies are damped but not to the extreme extent of the diffusive FTBS.

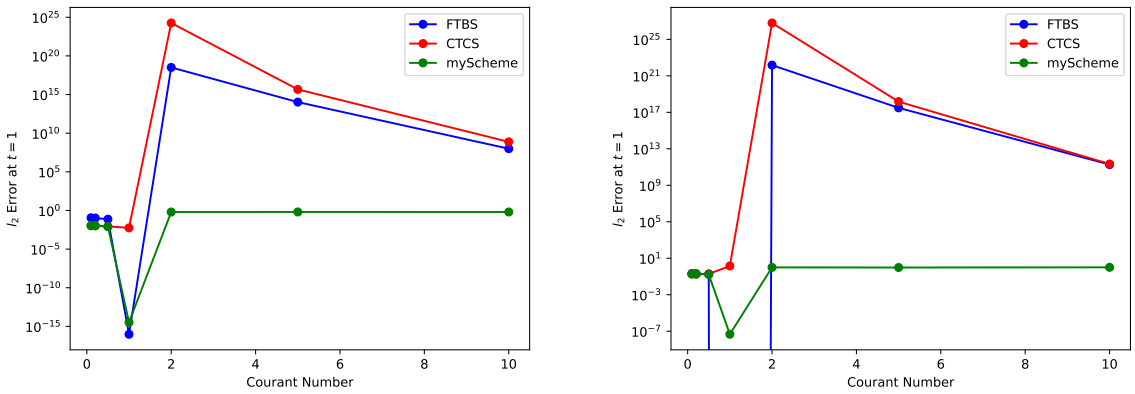


Figure 7: l_2 error at $t = 1$ for differing Δt with $\Delta x = 0.01$ for initial conditions (6) and (7) for the scheme defined in section 4.

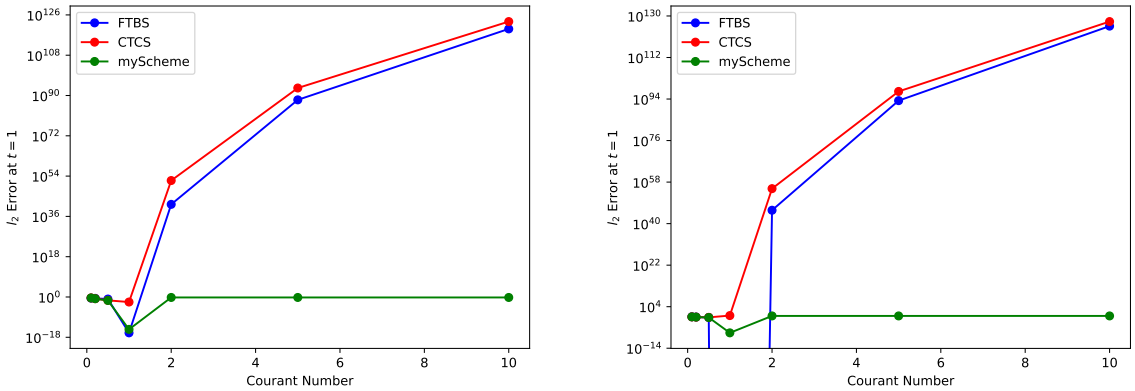


Figure 8: l_2 error at $t = 1$ for differing Δx with $\Delta t = 0.01$ for initial conditions (6) and (7) for the scheme defined in section 4.

5.4 Stability for Large Courant Numbers

We have also suggested that our new scheme will remain stable for large Courant numbers. To do so, we vary the Courant number and then find the l_2 error at $t = 1$. To ensure the method by which we vary c doesn't affect the outcome of our experiment, we will both vary Δt keeping Δx fixed (figure 7) and vary Δx keeping Δt fixed (figure 8).

In both figure 7 and 8, we see that the new scheme is unconditionally stable whereas the FTBS and CTCS schemes are divergent for $c > 1$, as described in Weller (2025).

Motivated by the interesting behaviour exhibited for small c, we repeat this experiment in figure 9 with $\Delta t = 0.01$.

Here, as predicted in section 4.3.2 that FTBS performs increasingly well as $c \to 1$ as the diffusion is minimised at this value. Moreover, it is interesting that our new scheme appears to display a similar behaviour for initial condition (7) (although this is not entirely clear from our plot).

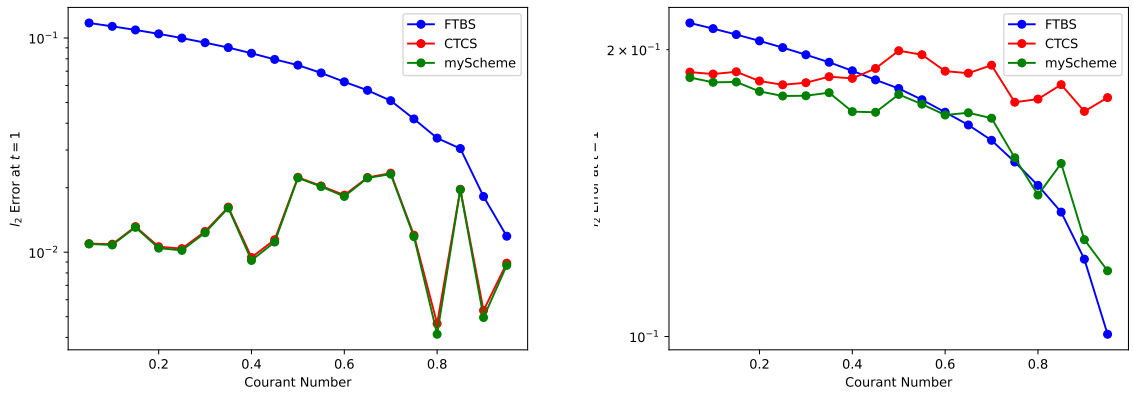


Figure 9: l_2 error at $t = 1$ for differing Δx with $\Delta t = 0.01$ for initial conditions (6) and (7) for the scheme defined in section 4.

6 Summary and Conclusions

6.1 Comparison of Schemes

In the above analysis, we have shown that CTCS and our new scheme perform fairly similarly in most respects, and so a major consideration is the storage usage and computational complexity of each scheme.

We expect N_t and N_x to be large in real world models, so we will use big-$\mathcal{O}$ notation. In general, each algorithm performs $\mathcal{O}(N_t \times N_x)$ steps to update ϕ using the stencil. Our new scheme performs N_x calculations to to find S_n at each step, therefore requiring an additional $\mathcal{O}(N_t \times N_x)$ steps in total. This is the same order as CTCS and so we can increase N_t and N_x without the complexity blowing up, but in practice this could mean a significant increase in the computation time. Similary, since CTCS uses two time-steps, the storage remains $\mathcal{O}(N_x)$, but increases by 50% which could practically be a large consideration. In short, practical limitations of computing power and memory size will play a significant role as to which scheme performs better.

6.2 Areas for Further Research

Our new scheme could be adapted in several ways:

- Depending on the quantity we are concerned with, we could choose our conserved S differently by simply changing the inner product. This would lead to entirely different schemes that might display different behaviour.
- When extending to a non-periodic region, we may no longer conserve S, but we may be able to evaluate the change in S at each time step by considering summing the fluxes across boundaries.
- We have chosen our stencil to include one time step and three spatial steps, but this was entirely arbitrary. In particular, for each additional spatial step, we gain an extra degree of freedom to either enforce consistency up to a higher degree or enforce another conserved quantity.

- This numerical scheme is highly non-linear as each coefficient is defined in terms of every value of the numerical solution in the previous time step. This means that much of the theory of numerical schemes cannot apply, meaning much more testing would be necessary to ensure the scheme is robust.

References

S. K. Godunov and I. Bohachevsky. Finite difference method for numerical computation of discontinuous solutions of the equations of fluid dynamics. 1959.

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